Supplementary material

Stage-specific process modelling and residual-learning analysis of grapevine phenology

This supplementary file reports supporting tables for the final main-analysis dataset, the fixed process–hybrid model comparison, and the final fixed model configuration used in the manuscript.

Table 1: Analysis-sample hierarchy used in the study.

Analysis set	Events	Sites	Site-years	Role in analysis
Spatial-validation source pool	31,439	2,324	17,172	Records with valid stage, coordinate and spatial-block information before process-model eligibility screening.
Process-model foundation	23,671	707	12,625	Process-model residual foundation before final residual-learning eligibility filter.
Main analysis sample	23,170	707	12,209	Final event set used for fixed process-vs-hybrid comparison, spatial robustness, ablation, and SHAP.

Table 2: Outer-fold performance of the selected process baseline and fixed hybrid model.

Stage	Fold	n	Process MAE	Process RMSE	Process bias	Hybrid MAE	Hybrid RMSE	Hybrid bias	ΔRMSE
BBCH07	1	743	6.4	8.8	0.2	6.2	8.0	-0.4	-0.8
	2	656	7.2	10.1	3.5	6.1	8.4	2.1	-1.7
	3	703	6.7	9.6	-0.7	6.2	8.6	-1.6	-1.0
	4	930	6.6	9.2	1.1	5.9	7.9	0.8	-1.2
	5	614	7.3	10.4	1.2	6.7	9.0	-1.3	-1.4
BBCH65	1	748	5.8	7.9	-0.5	5.0	7.4	-0.4	-0.5
	2	846	6.2	8.9	1.7	5.5	8.3	1.8	-0.6
	3	750	5.3	7.5	0.3	5.1	7.2	-0.9	-0.3
	4	1,133	5.4	8.0	-0.2	4.5	7.2	0.8	-0.8
	5	811	6.0	8.7	-0.6	5.4	8.5	-0.9	-0.2
BBCH81	1	700	13.2	16.4	-2.6	12.1	14.9	-1.4	-1.4
	2	913	10.5	13.2	3.6	10.3	12.7	2.1	-0.5
	3	753	14.2	17.4	-4.2	13.2	16.0	-2.3	-1.4
	4	1,116	11.8	14.9	-0.7	11.4	14.1	1.1	-0.8
	5	843	12.9	15.9	0.9	12.5	15.5	-0.9	-0.4
BBCH89	1	2,261	28.9	31.5	-28.2	8.3	10.5	-2.0	-21.0
	2	1,780	27.1	30.9	-25.9	9.2	12.4	-0.8	-18.5
	3	1,817	26.5	29.2	-25.6	7.9	10.1	-0.0	-19.1
	4	3,097	28.8	30.9	-28.5	7.4	9.5	2.5	-21.4
	5	1,956	26.6	29.6	-25.5	8.5	10.9	-1.3	-18.7

MAE, RMSE and bias are reported in days. ΔRMSE is hybrid minus process, so negative values indicate lower error after residual correction.

Table 3: Final fixed process and residual-learning model configuration.

Stage	Process model	Temperature branch	Calibrated process terms	Features	Residual learner	ML settings
BBCH07	BRIN-Aug1	Fold-specific selected branch	Chilling and forcing thresholds	11	Ridge	alpha = 0.01, 10, 100
BBCH65	GFV	Terrain-adjusted	Forcing threshold	12	RF	400 trees, depth 12, feature fraction 0.7, minimum leaf 10; 500 trees, depth 16, feature fraction 1.0, minimum leaf 10
BBCH81	GFV	Terrain-adjusted	Forcing threshold	12	Ridge	alpha = 100
BBCH89	PV-GDD10	Terrain-adjusted	Forcing threshold after predicted BBCH81	11	RF	300 trees, depth 8, feature fraction 0.7, minimum leaf 5